

FUNCTIONAL REQUIREMENTS DOCUMENT

AI-Powered Supply Chain & Procurement Analytics

Functional Requirements (FR) & Non-Functional Requirements (NFR)

Field	Detail
Project Name	AI-Powered Supply Chain & Procurement Analytics
Document Name	Functional & Non-Functional Requirements Document (FRD)
Version	1.0
Date	
Prepared By	Business Analyst
Document Status	Draft — Pending Stakeholder Review and Approval

Table of Contents

1. Document Information
2. Purpose of the FRD
3. Solution Overview
4. User Groups
5. Business Requirement Traceability
6. Functional Requirements
7. Non-Functional Requirements
8. Functional Workflow
9. Business Rules
10. Data Requirements
11. Dashboard & Reporting Requirements
12. Error & Exception Handling
13. Dependencies
14. Acceptance Considerations
15. Requirements Traceability
16. Future Functional Requirements

2. Purpose of the FRD

This Functional Requirements Document (FRD) defines the functional and non-functional capabilities required to deliver the AI-Powered Supply Chain & Procurement Analytics solution. It translates the approved Business Requirements Document (BRD) into requirements that are specific, measurable, and testable, and that can be used directly for development, Power BI implementation, AI/ML implementation, and User Acceptance Testing (UAT).

Each functional requirement in this document is traced back to an approved business requirement, and each functional area is supported by non-functional requirements that define the quality, reliability, and usability standards the solution must meet. Together, the FR and NFR sections provide a complete, self-contained specification for the solution; no separate NFR document is produced.

3. Solution Overview

The proposed solution is a centralized, AI-powered analytics capability built on the organization's historical supply-chain and procurement data (SupplyChain_Analytics_10000(2).csv, approximately 10,000 records spanning 2024-2025). The solution combines Excel, Power Query, SQL, Python, Power BI, and AI/ML techniques to prepare, validate, analyze, and report on supply-chain performance.

At a high level, the solution will:

- Prepare and validate raw transactional data into a consistent, documented analytical dataset.
- Provide performance monitoring across suppliers, order fulfillment, delivery/logistics, inventory, and cost/profitability.
- Calculate and report standardized KPIs through Power BI dashboards and recurring management reports.
- Apply AI/ML techniques to generate delivery-risk predictions and demand forecasts to support proactive decision-making.

The solution is scoped to the data currently available in the source dataset. Capabilities that depend on data or systems not present in the current dataset (e.g., real-time feeds, ERP integration) are documented separately in Section 16 as Future Requirements and are out of scope for the current project.

4. User Groups

The following user groups have been identified as consumers or administrators of the solution:

- Executive Sponsor — reviews executive-level summaries and overall supply-chain health.
- Supply Chain Manager — monitors end-to-end supply-chain performance and inventory risk.
- Procurement Manager — evaluates and compares supplier performance.
- Logistics Manager — analyzes delivery and transportation performance.
- Warehouse Manager — monitors inventory levels and warehouse-level performance.
- Finance Manager — analyzes cost and profitability.
- Operations Manager — monitors order fulfillment and operational status.
- Business Analyst — defines, documents, and maintains requirements, KPI definitions, and data definitions.
- Data Analyst — performs data validation, cleansing, and transformation.
- Data Scientist / ML Analyst — builds and maintains the delivery-risk and demand-forecasting models.
- BI Developer — builds and maintains Power BI dashboards and reports.

- Business Users — general consumers of dashboards and reports across the organization.

5. Business Requirement Traceability

The table below maps each approved business requirement to its related functional and non-functional requirements. Detailed requirement definitions follow in Sections 6 and 7.

BR ID	Business Requirement	Related FRs	Related NFRs
BR-001	Establish a validated, consistently defined dataset so that all downstream analysis and reporting is accurate and trustworthy.	FR-001–FR-007	NFR-003, NFR-006
BR-002	Provide visibility into overall supply-chain performance, including order volumes and order-status distribution, to support operational monitoring.	FR-008–FR-012, FR-045	NFR-001, NFR-004, NFR-005
BR-003	Enable evaluation and comparison of supplier performance to support supplier management decisions.	FR-013–FR-018	NFR-002, NFR-005
BR-004	Provide analysis of order fulfillment to identify delays and fulfillment trends across suppliers, products, and regions.	FR-019–FR-022	NFR-002, NFR-005
BR-005	Provide analysis of delivery and logistics performance across transportation modes and other dimensions to identify delay-prone areas.	FR-023–FR-027	NFR-002, NFR-004
BR-006	Provide visibility into inventory levels across products and warehouses to identify potential inventory-risk situations.	FR-028–FR-031	NFR-002, NFR-003
BR-007	Provide analysis of cost and profitability across suppliers, products, and other business dimensions to support financial decision-making.	FR-032–FR-034	NFR-002, NFR-005
BR-008	Provide standardized KPI monitoring and management reporting (operational, weekly, monthly, and executive) built on documented KPI definitions.	FR-035–FR-038, FR-037a	NFR-001, NFR-002, NFR-005, NFR-009, NFR-010
BR-009	Provide a data-driven indication of delivery risk to allow proactive management of at-risk orders.	FR-039–FR-041	NFR-007, NFR-008
BR-010	Provide demand forecasting based on historical order data to support planning decisions.	FR-042–FR-044	NFR-007, NFR-008

6. Functional Requirements

All functional requirements below are written using "shall" language and are traceable to an approved business requirement (Section 5). Requirements are grouped by functional category (A–M) as defined in the project scope.

A. Data Preparation & Validation

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-001	BR-001	The system shall standardize date fields (Order Date, Shipment Date, Delivery Date) into a consistent format prior to analysis.	Data Analyst / BI Developer	High	Raw date fields from SupplyChain_Analytics_10000(2).csv	Parse and convert raw date values into a single standardized date format; flag unparseable values.	Standardized date fields ready for analysis	Ensures accurate time-based analysis and trend reporting.	AC-DP-01
FR-002	BR-001	The system shall validate that required fields (Order Date, Customer, Product, Supplier, Order Quantity, Unit Price, Order Status) are present for each record.	Data Analyst	High	Raw dataset records	Check each record for presence of required fields against a documented list.	List of records passing/failing required-field validation	Prevents incomplete records from distorting analysis.	AC-DP-02
FR-003	BR-001	The system shall identify records with missing values in key analytical fields.	Data Analyst	High	Validated dataset	Scan key fields for nulls/blanks and log the record and field affected.	Missing-value log	Supports transparent data-quality reporting.	AC-DP-03
FR-004	BR-001	The system shall identify records with invalid values (e.g., negative Order Quantity, negative Unit Price, unrecognized Order Status).	Data Analyst	High	Validated dataset	Apply documented business rules to flag records outside acceptable ranges/values.	Invalid-record log	Prevents invalid data from affecting KPIs and models.	AC-DP-04
FR-005	BR-001	The system shall identify duplicate records within the dataset.	Data Analyst	Medium	Validated dataset	Compare records on key identifying fields to detect duplicates.	Duplicate-record log	Avoids double-counting in KPI and volume metrics.	AC-DP-05
FR-006	BR-001	The system shall apply approved data transformations (e.g., standardized categories, calculated fields) using Power Query, SQL, and/or Python.	BI Developer / Data Analyst	High	Cleaned dataset	Execute documented transformation logic to prepare the analytical dataset.	Transformed analytical dataset	Produces a consistent dataset for reporting and modeling.	AC-DP-06
FR-007	BR-001	The system shall maintain documented data definitions for all fields used in analysis and reporting.	Business Analyst	Medium	Dataset field list	Maintain a data dictionary describing each field, its source, and its meaning.	Data dictionary document	Ensures consistent interpretation of data across users.	AC-DP-07

B. Supply-Chain Performance

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-008	BR-002	The system shall display the total number of orders for a selected period.	Supply Chain Manager / Operations Manager	High	Transformed order data	Aggregate order count by selected filters/date range.	Total order count	Provides a baseline operational volume metric.	AC-SCP-01
FR-009	BR-002	The system shall display the distribution of orders by Order Status (Delivered, In Transit, Delayed, Cancelled).	Supply Chain Manager	High	Transformed order data	Group and count orders by Order Status value.	Order-status distribution visual	Highlights operational health at a glance.	AC-SCP-02
FR-010	BR-002	The system shall allow users to monitor Delivered, Delayed, Cancelled, and In-Transit order counts and trends.	Operations Manager	High	Transformed order data	Calculate counts/trends per status category over time.	Status monitoring visuals	Enables early identification of operational issues.	AC-SCP-03
FR-011	BR-002	The system shall allow users to analyze supply-chain performance trends over time (monthly/quarterly).	Supply Chain Manager	Medium	Transformed order data with standardized dates	Aggregate metrics by time period and render trend visuals.	Time-series performance visuals	Supports identification of seasonal or emerging patterns.	AC-SCP-04
FR-012	BR-002	The system shall allow users to filter supply-chain performance views by Region, Product Category, Supplier, and Transportation Mode.	Business User	High	Transformed order data	Apply user-selected filters to underlying dataset and refresh visuals.	Filtered performance visuals	Allows targeted operational analysis.	AC-SCP-05

C. Supplier Analysis

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-013	BR-003	The system shall allow users to compare suppliers on order volume, delays, cancellations, cost, and profitability.	Procurement Manager	High	Transformed order data by Supplier	Aggregate key metrics by Supplier and present side-by-side comparison.	Supplier comparison visual/table	Supports supplier evaluation decisions.	AC-SUP-01
FR-014	BR-003	The system shall display order volume by Supplier.	Procurement Manager	Medium	Transformed order data	Aggregate order counts grouped by Supplier.	Supplier order-volume visual	Identifies most/least utilized suppliers.	AC-SUP-02
FR-015	BR-003	The system shall calculate and display the rate of delayed orders by Supplier.	Procurement Manager	High	Transformed order data (Order Status, Delivery Time)	Calculate delayed-order percentage grouped by Supplier.	Supplier delay-rate visual	Flags suppliers contributing to delivery risk.	AC-SUP-03

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-016	BR-003	The system shall calculate and display the rate of cancelled orders by Supplier.	Procurement Manager	Medium	Transformed order data (Order Status)	Calculate cancellation percentage grouped by Supplier.	Supplier cancellation-rate visual	Identifies suppliers with fulfillment reliability concerns.	AC-SUP-04
FR-017	BR-003	The system shall display total cost, shipping cost, and profit associated with each Supplier.	Finance Manager / Procurement Manager	High	Transformed order data (Total Cost, Shipping Cost, Profit)	Aggregate cost and profit fields grouped by Supplier.	Supplier cost/profitability visual	Supports cost-driven supplier decisions.	AC-SUP-05
FR-018	BR-003	The system shall identify suppliers that meet a documented underperformance definition (e.g., delay rate and/or cancellation rate above an agreed threshold).	Procurement Manager	Medium	Supplier-level aggregated metrics	Apply documented underperformance criteria to supplier metrics.	Underperforming-supplier list	Enables proactive supplier management.	AC-SUP-06

D. Order Fulfillment

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-019	BR-004	The system shall display order counts and percentages by Order Status.	Operations Manager	High	Transformed order data	Aggregate and calculate percentage share per Order Status.	Order status summary visual	Provides a fulfillment-health overview.	AC-OF-01
FR-020	BR-004	The system shall identify orders with a Delayed status and their associated attributes (Supplier, Product, Region, Transportation Mode).	Operations Manager	High	Transformed order data	Filter dataset for Delayed status and present associated attributes.	Delayed-order list/visual	Enables root-cause review of delays.	AC-OF-02
FR-021	BR-004	The system shall display fulfillment trends over time.	Operations Manager	Medium	Transformed order data with standardized dates	Aggregate fulfillment metrics by time period.	Fulfillment trend visual	Identifies improving or worsening fulfillment patterns.	AC-OF-03
FR-022	BR-004	The system shall allow users to analyze fulfillment performance by Supplier, Product, and Region where supported by the data.	Business User	Medium	Transformed order data	Aggregate fulfillment metrics grouped by the selected dimension.	Dimensional fulfillment visuals	Supports targeted fulfillment analysis.	AC-OF-04

E. Delivery & Logistics

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-023	BR-005	The system shall calculate and display average and distribution of Delivery Time.	Logistics Manager	High	Transformed order data (Delivery Time)	Calculate average/median/distribution of Delivery Time.	Delivery-time visual	Establishes delivery performance baseline.	AC-DL-01

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-024	BR-005	The system shall allow comparison of delivery performance across Transportation Modes.	Logistics Manager	High	Transformed order data (Transportation Mode, Delivery Time, Order Status)	Aggregate delivery metrics grouped by Transportation Mode.	Transportation-mode comparison visual	Supports logistics mode selection decisions.	AC-DL-02
FR-025	BR-005	The system shall compare delivery performance across Warehouses and Regions.	Logistics Manager / Warehouse Manager	Medium	Transformed order data	Aggregate delivery metrics grouped by Warehouse/Region.	Warehouse/region delivery comparison visual	Identifies logistics bottlenecks by location.	AC-DL-03
FR-026	BR-005	The system shall identify Regions, Warehouses, Suppliers, or Transportation Modes associated with higher delay rates.	Logistics Manager	High	Transformed order data	Calculate delay rate per dimension and rank/flag highest values.	High-delay-area visual/list	Enables targeted logistics interventions.	AC-DL-04
FR-027	BR-005	The system shall allow users to analyze delivery performance by Supplier, Product, Warehouse, Region, and Transportation Mode where supported by the data.	Business User	Medium	Transformed order data	Aggregate delivery metrics grouped by the selected dimension(s).	Dimensional delivery-performance visuals	Provides flexible logistics analysis.	AC-DL-05

F. Inventory Analysis

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-028	BR-006	The system shall display current Inventory Level by Product.	Warehouse Manager	High	Transformed inventory data (Inventory Level)	Aggregate/display Inventory Level grouped by Product.	Inventory-by-product visual	Provides inventory visibility for planning.	AC-INV-01
FR-029	BR-006	The system shall allow comparison of Inventory Level across Products and Warehouses.	Warehouse Manager	High	Transformed inventory data	Aggregate Inventory Level grouped by Product and Warehouse.	Product/warehouse inventory comparison visual	Supports stock balancing decisions.	AC-INV-02
FR-030	BR-006	The system shall identify potential inventory-risk situations using a documented, data-supported definition (e.g., inventory level below a defined analytical threshold derived from historical data).	Warehouse Manager / Supply Chain Manager	Medium	Transformed inventory data	Apply documented inventory-risk logic to Inventory Level values.	Inventory-risk flag/visual	Enables proactive stock management.	AC-INV-03
FR-031	BR-006	The system shall display Inventory Level trends over time.	Warehouse Manager	Medium	Transformed inventory data with standardized dates	Aggregate Inventory Level by time period.	Inventory trend visual	Identifies stock depletion/accumulation patterns.	AC-INV-04

G. Cost & Profitability

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-032	BR-007	The system shall calculate and display Total Cost, Unit Price, Shipping Cost, and Profit summary metrics.	Finance Manager	High	Transformed order data (Total Cost, Unit Price, Shipping Cost, Profit)	Aggregate financial fields for the selected filter context.	Cost/profit summary visual	Establishes financial performance baseline.	AC-COST-01
FR-033	BR-007	The system shall allow comparison of cost by Supplier, Product, and Product Category.	Finance Manager / Procurement Manager	High	Transformed order data	Aggregate cost fields grouped by the selected dimension.	Cost comparison visual	Supports cost-optimization decisions.	AC-COST-02
FR-034	BR-007	The system shall allow comparison of profitability across Supplier, Product, Region, and other relevant dimensions.	Finance Manager	High	Transformed order data (Profit)	Aggregate Profit grouped by the selected dimension.	Profitability comparison visual	Identifies most/least profitable segments.	AC-COST-03

H. KPI Management

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-035	BR-008	The system shall calculate approved KPIs (e.g., On-Time Delivery Rate, Delay Rate, Cancellation Rate, Average Delivery Time, Total Profit Margin) using documented definitions.	BI Developer / Business Analyst	High	Transformed dataset and documented KPI definitions	Apply documented formulas to calculate each KPI.	KPI values	Provides standardized performance indicators.	AC-KPI-01
FR-036	BR-008	The system shall display KPI trends over time and allow filtering by relevant dimensions.	Business User	High	Calculated KPI values	Aggregate KPI values by time period and apply user filters.	KPI trend visuals	Enables monitoring of performance over time.	AC-KPI-02

I. Power BI Reporting

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-037	BR-008	The system shall provide Power BI dashboards covering Executive Overview, Supplier Analysis, Logistics/Delivery, Inventory, Cost/Profitability, and Risk Analysis.	BI Developer	High	Transformed dataset, calculated KPIs, model outputs	Build and publish Power BI report pages per defined dashboard scope.	Published Power BI dashboards	Provides self-service analytical access for business users.	AC-BI-01

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-037a	BR-008	Users shall be able to filter, drill down where applicable, compare dimensions, view KPI trends, and identify exceptions within Power BI dashboards.	Business User	High	Published Power BI dashboards	Provide interactive filter, drill-down, and comparison controls.	Interactive dashboard views	Enables self-service exploration of the data.	AC-BI-02

J. Management Reporting

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-038	BR-008	The system shall support operational, Weekly Business Review (WBR), Monthly Business Review (MBR), and executive summary reporting, highlighting significant trends, risks, and business issues.	Executive Sponsor / Operations Manager	High	Calculated KPIs and dashboard outputs	Compile relevant KPIs and highlights into the applicable reporting cadence/format.	Operational / WBR / MBR / executive reports	Supports management decision-making at the appropriate cadence.	AC-MGT-01

K. Delivery-Risk Prediction

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-039	BR-009	The system shall use approved historical order data to train and apply a delivery-risk prediction model.	Data Scientist / ML Analyst	High	Approved historical order dataset	Train and apply a documented predictive model to estimate delivery risk.	Delivery-risk predictions	Provides a data-driven basis for proactive risk management.	AC-RISK-01
FR-040	BR-009	The system shall produce and display a documented risk score or risk category for applicable orders/records.	Logistics Manager / Supply Chain Manager	High	Model output	Map model output to a documented risk score/category and present in reporting.	Risk score/category visual	Allows users to identify higher-risk orders for review.	AC-RISK-02
FR-041	BR-009	The system shall display model performance metrics (e.g., accuracy, precision/recall, or an equivalent documented metric) for the delivery-risk model.	Data Scientist / ML Analyst	Medium	Model evaluation results	Calculate and present documented evaluation metrics.	Model performance summary	Supports transparency in model reliability.	AC-RISK-03

L. Demand Forecasting

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-042	BR-010	The system shall generate demand forecasts using historical Order Quantity data at an approved level (e.g., Product / Category / Month).	Data Scientist / ML Analyst	High	Historical Order Quantity data	Apply a documented forecasting method to generate forward-looking demand estimates.	Demand forecast values	Supports planning and procurement decisions.	AC-FCST-01
FR-043	BR-010	The system shall display forecast results and document the forecasting methodology used.	Supply Chain Manager	Medium	Forecast output	Present forecast results alongside a description of the methodology.	Forecast visual and methodology note	Ensures forecast transparency for business users.	AC-FCST-02
FR-044	BR-010	The system shall provide appropriate forecast evaluation metrics (e.g., MAPE, MAE, or an equivalent documented metric).	Data Scientist / ML Analyst	Medium	Forecast output vs. historical actuals	Calculate documented forecast accuracy metrics.	Forecast evaluation summary	Supports confidence assessment in forecast outputs.	AC-FCST-03

M. User Interaction

FR ID	BR ID	Functional Requirement	User/Actor	Priority	Input	Functional Behavior	Output	Business Purpose	Acceptance Ref.
FR-045	BR-002	Users shall be able to filter reports, view relevant record-level detail, navigate between analytical sections, and interpret KPI and risk information using consistent terminology.	Business User	High	Published dashboards and reports	Provide consistent navigation, filtering, and terminology across all report pages.	Consistent, navigable user experience	Ensures the solution is usable by non-technical business users.	AC-UI-01

7. Non-Functional Requirements

Non-functional requirements define the quality attributes the solution must meet in addition to its functional capabilities. Where a measurable threshold has not yet been agreed with stakeholders, it is explicitly marked "To Be Agreed" rather than an invented value.

NFR ID	BR/FR Ref.	Category	Non-Functional Requirement	Measurement / Acceptance Standard	Priority
NFR-001	FR-037, FR-045	Usability	The dashboard shall use clear and consistent terminology understandable to business users.	Terminology reviewed and confirmed by business stakeholders during UAT.	High
NFR-002	FR-035, FR-036	Consistency	All KPI calculations shall use documented definitions and produce consistent results across reports.	KPI values match documented formulas across all report pages during UAT verification.	High
NFR-003	FR-003, FR-004, FR-005	Data Quality	Data-quality issues identified during preparation shall be documented and traceable.	100% of logged missing/invalid/duplicate records are captured in the data-quality log.	High
NFR-004	FR-037, FR-037a	Performance	Dashboard interactions shall respond within an acceptable business-user response time under normal usage.	To Be Agreed (target response time to be confirmed with stakeholders).	Medium
NFR-005	FR-035, FR-036, FR-037a	Reliability	The solution shall maintain consistent KPI calculations across report pages.	No variance in KPI totals when the same filter context is applied across pages, verified during UAT.	High
NFR-006	FR-006, FR-007	Maintainability	The solution shall document data transformations sufficiently to support maintenance and auditability.	Transformation logic documented for 100% of transformation steps used in the analytical dataset.	Medium
NFR-007	FR-039, FR-041	Model Reliability / Transparency	Predictive models shall report appropriate evaluation metrics.	Delivery-risk model publishes at least one documented performance metric.	High
NFR-008	FR-039, FR-040	Model Reliability / Transparency	Predictive outputs shall identify the model's intended purpose and limitations.	Risk and forecast outputs include a documented purpose/limitations statement.	High
NFR-009	FR-037, FR-038	Security	Access to sensitive business reporting should follow appropriate user permissions where implemented.	To Be Agreed (role-based access model to be confirmed with IT/security stakeholders).	Medium
NFR-010	FR-006, FR-007, FR-035	Scalability / Maintainability	The analytical solution should be maintainable so that KPI definitions, data transformations, and reports can be updated without redesigning the entire solution.	Updates to a single KPI or transformation do not require rebuilding unrelated report pages, verified during change testing.	Medium

8. Functional Workflow

The solution follows a consistent end-to-end flow, from raw data input through to business action:

Data Input → Data Validation → Data Transformation → Data Analysis → KPI Calculation → Dashboard Reporting → Risk Prediction / Forecasting → Business Insight → Management Decision → Business Action

1. Data Input

Raw supply-chain and procurement data (SupplyChain_Analytics_10000(2).csv) is loaded into the analytical environment.

2. Data Validation

Required fields, missing values, invalid records, and duplicates are identified against documented rules (Section 9).

3. Data Transformation

Approved transformations are applied using Power Query, SQL, and/or Python to produce a clean analytical dataset.

4. Data Analysis

Business users and analysts explore supplier, fulfillment, delivery, inventory, and cost/profitability data.

5. KPI Calculation

Approved KPIs are calculated using documented definitions and made available for reporting.

6. Dashboard Reporting

Power BI dashboards present KPIs, trends, and comparisons across the six defined dashboard pages.

7. Risk Prediction / Forecasting

AI/ML models generate delivery-risk indicators and demand forecasts from approved historical data.

8. Business Insight

Users interpret dashboard outputs, KPI trends, and predictive indicators to identify issues and opportunities.

9. Management Decision

Managers and executives use insights and recurring reports to make operational and strategic decisions.

10. Business Action

Decisions are translated into operational actions (e.g., supplier follow-up, inventory reallocation, logistics adjustments).

9. Business Rules

The following business rules govern data handling and analytical processing within the solution. These rules are limited to what is supported by the dataset and the approved BRD; no procurement policies beyond the dataset's scope are assumed.

- Order status must use one of the approved values: Delivered, In Transit, Delayed, or Cancelled.
- Order quantity must be greater than zero to be treated as a valid order record.
- Negative unit price values must be treated as invalid unless explicitly justified and documented.
- Dates must be standardized to a single format before any time-based analysis is performed.
- KPI definitions must be documented and approved before being used in reporting.
- Derived metrics must have documented formulas maintained in the data dictionary.

- Predictive models (delivery-risk prediction and demand forecasting) must use only approved historical data.
- Invalid or missing records must be identified and handled per the data-quality process before being used in analysis.
- Duplicate records must be identified and resolved before being included in aggregate metrics.
- Any threshold used to classify inventory risk or supplier underperformance must be documented and based on available data.

10. Data Requirements

The table below documents the dataset fields required to support the functional requirements defined in Section 6. Only fields available in the source dataset are used; no unavailable fields (e.g., Purchase Order Number, Supplier Contract Value) are assumed.

Data Req. ID	Related FR	Data Field	Purpose	Source	Transformation	Validation Rule	Output / Usage
DR-001	FR-001, FR-011	Order Date	Basis for time-series and trend analysis	SupplyChain_Analytics_10000(2).csv	Standardize to a single date format	Must be a valid, parseable date	Used in performance, fulfillment, and trend visuals
DR-002	FR-023, FR-024	Shipment Date / Delivery Date	Basis for delivery-time calculation	SupplyChain_Analytics_10000(2).csv	Standardize to a single date format	Delivery Date must not precede Shipment Date	Used in delivery-time and logistics visuals
DR-003	FR-019, FR-020	Order Status	Classification of order outcome	SupplyChain_Analytics_10000(2).csv	Standardize category labels	Must be one of: Delivered, In Transit, Delayed, Cancelled	Used in status distribution and fulfillment visuals
DR-004	FR-013, FR-014	Supplier / Supplier Region	Supplier-level grouping and comparison	SupplyChain_Analytics_10000(2).csv	Standardize naming, remove duplicates	Must not be blank	Used in supplier analysis visuals
DR-005	FR-024	Transportation Mode	Logistics-mode comparison	SupplyChain_Analytics_10000(2).csv	Standardize category labels	Must be a recognized mode value	Used in delivery/logistics visuals
DR-006	FR-008, FR-032	Order Quantity / Unit Price	Volume and revenue-related calculations	SupplyChain_Analytics_10000(2).csv	None beyond validation	Order Quantity > 0; Unit Price >= 0	Used in performance and cost visuals
DR-007	FR-032, FR-033	Total Cost / Shipping Cost / Profit	Cost and profitability analysis	SupplyChain_Analytics_10000(2).csv	None beyond validation	Must be numeric; flagged if negative without justification	Used in cost/profitability visuals
DR-008	FR-028, FR-030	Inventory Level	Inventory monitoring and risk identification	SupplyChain_Analytics_10000(2).csv	None beyond validation	Must be numeric and non-negative	Used in inventory visuals
DR-009	FR-012, FR-025	Customer Region / Country / Warehouse	Dimensional filtering and comparison	SupplyChain_Analytics_10000(2).csv	Standardize naming	Must not be blank	Used as filter dimensions across dashboards
DR-010	FR-042, FR-039	Order Quantity, Order Date, Order Status (historical series)	Model input for forecasting and delivery-risk prediction	SupplyChain_Analytics_10000(2).csv (approved historical subset)	Aggregate to model-required granularity	Sufficient historical observations confirmed prior to modeling	Used as input to forecasting and risk-prediction models

11. Dashboard & Reporting Requirements

The solution will deliver six Power BI dashboard pages. Detailed UI wireframes are outside the scope of this FRD; the table below defines the business purpose, target users, and required content for each page.

Dashboard	Business Purpose	Target User	Required KPIs	Required Filters	Required Visuals	Expected Business Decision
Executive Dashboard	Provide a high-level view of overall supply-chain health for leadership.	Executive Sponsor, Operations Manager	Total Orders, On-Time Delivery Rate, Delay Rate, Cancellation Rate, Total Profit	Date Range, Region, Product Category	KPI cards, trend lines, status distribution	Identify areas requiring management attention
Supplier Dashboard	Support supplier evaluation and comparison.	Procurement Manager, Business Analyst	Supplier Delay Rate, Cancellation Rate, Order Volume, Supplier Cost/Profit	Supplier, Supplier Region, Date Range	Ranked bar charts, comparison tables	Identify underperforming suppliers and renegotiation candidates
Delivery & Logistics Dashboard	Analyze delivery performance and transportation efficiency.	Logistics Manager, Warehouse Manager	Average Delivery Time, Delay Rate by Mode/Region, On-Time Delivery Rate	Transportation Mode, Warehouse, Region, Date Range	Trend lines, comparison charts, delay heat view	Identify logistics bottlenecks and mode optimization opportunities
Inventory Dashboard	Monitor inventory positions and flag risk situations.	Warehouse Manager, Supply Chain Manager	Inventory Level by Product/Warehouse, Inventory-Risk Flags	Product, Warehouse, Date Range	Trend lines, comparison bars, risk indicators	Prioritize replenishment or reallocation actions
Cost & Profitability Dashboard	Analyze cost drivers and profitability across dimensions.	Finance Manager, Procurement Manager	Total Cost, Shipping Cost, Profit, Profit Margin	Supplier, Product Category, Region, Date Range	Trend lines, comparison charts, profitability tables	Identify cost-reduction and profitability-improvement opportunities
Risk / Forecast Dashboard	Present delivery-risk predictions and demand forecasts.	Supply Chain Manager, Data Scientist / ML Analyst	Delivery-Risk Score/Category, Forecasted Demand, Model Performance Metrics	Product, Category, Region, Date Range	Risk indicators, forecast trend lines, model-metric cards	Prioritize at-risk orders and inform demand planning

12. Error & Exception Handling

The following table defines expected system behavior when data-quality or predictive-output exceptions are encountered.

Error / Exception	Expected Behavior	Business Impact	Resolution / Handling
Missing data in a required field	Record is flagged and logged; excluded from KPI calculations until resolved or documented as accepted.	Potential understatement of volume-based metrics if unresolved.	Reviewed by Data Analyst; corrected at source or documented as an accepted exception.
Duplicate records	Duplicate records are flagged and excluded from aggregate counts.	Risk of inflated order/volume metrics if not handled.	Reviewed by Data Analyst; de-duplicated using documented matching logic.
Invalid dates	Record is flagged; excluded from time-based analysis until corrected.	Potential gaps or distortion in trend analysis.	Reviewed and corrected at source, or documented as an accepted exception.
Invalid quantities (e.g., Order Quantity <= 0)	Record is flagged as invalid and excluded from volume/cost calculations.	Prevents distortion of order-volume and cost metrics.	Reviewed by Data Analyst; corrected at source or documented as an accepted exception.
Missing supplier information	Record is flagged; excluded from supplier-level analysis.	Understates supplier-level metrics if unresolved.	Reviewed by Data Analyst; corrected at source or documented as an accepted exception.
Missing product information	Record is flagged; excluded from product-level analysis.	Understates product-level metrics if unresolved.	Reviewed by Data Analyst; corrected at source or documented as an accepted exception.
Inconsistent category values (e.g., inconsistent Region naming)	Values are standardized using documented mapping rules during transformation.	Prevents fragmented or duplicated dimensional groupings.	Standardized during the transformation step; mapping documented in the data dictionary.
Unsupported or low-confidence predictive outputs	Output is withheld or clearly labeled as low-confidence; not presented as a guaranteed result.	Prevents overreliance on unreliable predictions.	Reviewed by Data Scientist / ML Analyst; model retrained or output suppressed as appropriate.

13. Dependencies

Delivery of this solution depends on the following:

- Availability of the SupplyChain_Analytics_10000(2).csv dataset for the full analysis period (2024-2025).
- Data quality of the source dataset is sufficient to support validation and transformation.
- Approved and documented KPI definitions are available prior to KPI implementation.
- Stakeholder validation and sign-off on business rules, KPI definitions, and dashboard requirements.
- Availability of sufficient historical data to support delivery-risk prediction and demand forecasting.
- Sufficient historical observations exist to support reliable AI/ML model training and evaluation.
- Availability of a Power BI environment for dashboard development and publishing.
- Availability of a SQL/Python processing environment for data preparation and modeling.

14. Acceptance Considerations

For each major functional area, the following must be true for the functionality to be considered complete and ready for sign-off:

Functional Area	Completion Criteria
Data Preparation	All required fields validated; missing, invalid, and duplicate records identified and logged; documented data dictionary available.
Supplier Analysis	Supplier comparison, delay rate, cancellation rate, cost, and profitability visuals produce results consistent with the validated dataset.
Order Fulfillment	Order-status summaries and delayed-order identification match validated dataset totals.
Delivery Analysis	Delivery-time and transportation-mode comparisons are accurate and consistent with the validated dataset.
Inventory Analysis	Inventory-level visuals and documented risk flags reflect the validated dataset and approved risk definition.
Cost Analysis	Cost and profitability visuals reconcile to validated Total Cost, Shipping Cost, and Profit fields.
KPI Reporting	All approved KPIs calculate consistently across report pages using documented definitions.
Power BI	All six dashboards are published, filterable, and reviewed/approved by target business users.
Delivery Risk Prediction	Risk model produces documented risk scores/categories and reports evaluation metrics; outputs labeled as predictive, not guaranteed.
Demand Forecasting	Forecast output is produced at the approved level with documented methodology and evaluation metrics.

15. Requirements Traceability

The table below provides an end-to-end traceability sample linking business requirements through to functional requirements, non-functional requirements, user stories, and acceptance criteria. This structure is applied consistently across the full requirement set documented in Sections 5-7.

BR ID	FR ID	NFR ID	User Story ID	Acceptance Criteria ID
BR-001	FR-001	NFR-003	US-001	AC-DP-01
BR-001	FR-002	NFR-003	US-002	AC-DP-02
BR-001	FR-004	NFR-003	US-003	AC-DP-04
BR-002	FR-008	NFR-004	US-004	AC-SCP-01
BR-002	FR-009	NFR-001	US-005	AC-SCP-02
BR-003	FR-013	NFR-002	US-006	AC-SUP-01
BR-003	FR-018	NFR-005	US-007	AC-SUP-06
BR-004	FR-020	NFR-002	US-008	AC-OF-02
BR-005	FR-024	NFR-004	US-009	AC-DL-02
BR-005	FR-026	NFR-002	US-010	AC-DL-04
BR-006	FR-030	NFR-003	US-011	AC-INV-03
BR-007	FR-032	NFR-005	US-012	AC-COST-01
BR-008	FR-035	NFR-002	US-013	AC-KPI-01
BR-008	FR-037	NFR-009	US-014	AC-BI-01
BR-008	FR-038	NFR-001	US-015	AC-MGT-01
BR-009	FR-039	NFR-007	US-016	AC-RISK-01
BR-009	FR-040	NFR-008	US-017	AC-RISK-02

BR ID	FR ID	NFR ID	User Story ID	Acceptance Criteria ID
BR-010	FR-042	NFR-007	US-018	AC-FCST-01
BR-010	FR-043	NFR-008	US-019	AC-FCST-02
BR-002	FR-045	NFR-001	US-020	AC-UI-01

16. Future Functional Requirements

The following capabilities are identified as potential future enhancements. They are explicitly out of scope for the current project and are not reflected in the functional or non-functional requirements above.

- Real-time supply-chain monitoring (current scope is based on historical/batch data).
- ERP system integration for direct data feeds.
- Automated alerting/notifications for risk or exception conditions.
- Real-time inventory feeds (current scope uses dataset-provided inventory snapshots).
- Supplier portal integration for direct supplier data exchange.
- Automated procurement workflows (e.g., automated purchase order generation).